Preliminary transcriptomic impressions

This is a descriptive analysis of the independently QC-filtered GSE290989 data, not a reproduction of the authors' annotated atlas and not a formal differential-expression analysis.

Cardiomyocytes (46,485 provisionally labeled nuclei): The clearest impression is a developmental transition toward contractile and excitation-contraction maturity. Tnni1 falls strongly with age, Myh6 increases, and the aggregate calcium/ion-handling panel is higher at P14-P21 than at P0. Atp2a2 rises from about 2,863 to 4,959 pseudobulk CPM between P0 and P21. Individual calcium genes do not all rise monotonically: Ryr2 peaks around P14. Cell-cycle genes are higher at P0/P7 than P14/P21. The fatty-acid/oxidative panel peaks at P7 and individual metabolic genes have different trajectories; a simple monotonic metabolic maturation claim is not supported by this panel.

Fibroblasts (14,860 provisionally labeled nuclei): Early postnatal profiles emphasize proliferation and developmental extracellular-matrix remodeling. Postn, Tnc and Fn1 are higher at P0/P7 and decline later. Fibrillar collagen expression peaks around P7: Col1a1 and Col1a2 then decline, while Dcn remains relatively high. Pi16 rises at P14/P21 compared with P0, consistent with changes in interstitial fibroblast states. This is developmental remodeling; these markers alone do not establish pathological fibrosis.

Endothelial cells (4,956 provisionally labeled nuclei): Early profiles have higher proliferative/angiogenic activity, followed by stronger differentiated vascular signatures. Mki67 and Top2a peak at P7 and fall sharply by P21; Apln is elevated at P7/P14. Capillary-associated Car4 and Cd36 increase with age. Arterial and venous marker panels are also higher later, but this can reflect changing subtype composition as well as within-subtype maturation. Lymphatic markers are inconsistent between replicates and do not justify a strong temporal conclusion.

Cross-lineage impression: The profiles are compatible with coordinated postnatal maturation: early growth and matrix/vascular remodeling give way to stronger functional cardiac and vascular specialization. They do not yet identify causal intercellular signals or distinguish subtype redistribution from within-cell-state change.

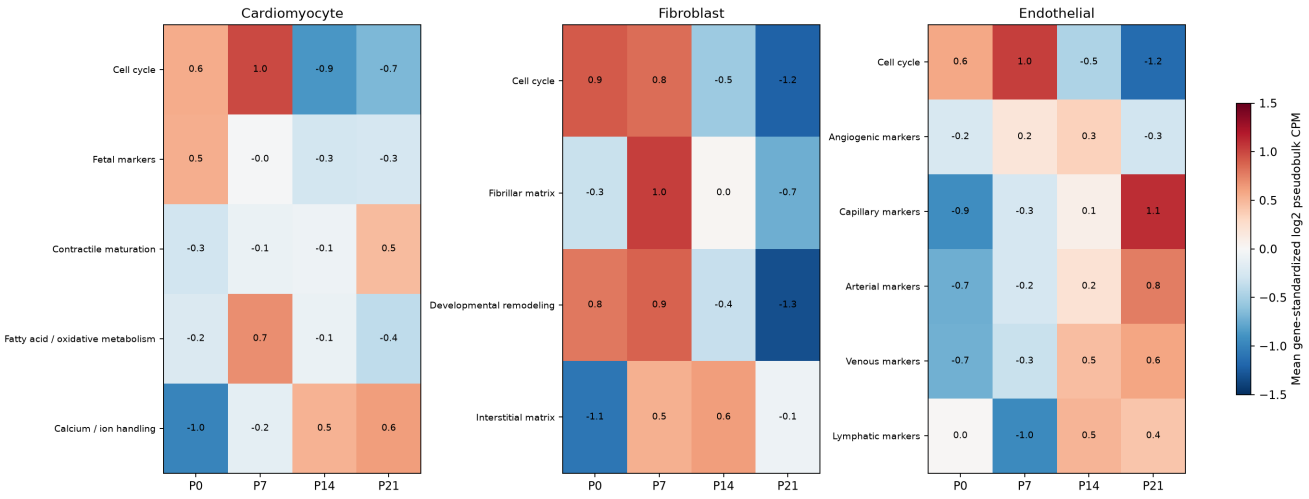
Methods: The shared QC neighbor graph was clustered with Leiden resolution 0.8 (seed 42). Broad lineage labels were assigned from multi-gene canonical marker panels, including alternative lineages (pericyte, smooth muscle, immune, epicardial, glial). A cluster required at least two supported markers, top standardized marker score >0.2 and a score margin >0.2; 1,011 nuclei remain uncertain/mixed. These are preliminary cluster labels, not independently validated cell identities. Canonical lineage genes and complete annotation diagnostics are saved with the analysis.

Expression was aggregated within each lineage and library, normalized by total lineage/library UMIs to CPM, and summarized equally across the two libraries at each age. Program heatmaps average gene-standardized log2(CPM+1) across predefined panels; colors compare a program over time, not its absolute strength against another program. Raw-count and corrected-count program profiles correlate at r=0.987-1.000 across the eight libraries; this is a correction sensitivity check, not independent biological validation. Replicate program scores are provided separately.

Limits: No significance tests or causal claims are made. Broad labels need review using unbiased cluster markers and reference annotations. Lineage mixtures, technical effects, nuclear RNA sampling, and differential QC retention can influence the profiles. P7 retained fewer nuclei after QC, especially among some non-cardiomyocyte clusters. Nucleus proportions are not tissue-cell proportions. Animal/pooling and capture provenance still require confirmation before replicate-level

inference. Xenium data have not been used to confirm localization of these states.

Temporal expression programs



Preliminary broad lineage assignments

